Deep fake Video Detection and Face Detection

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Abstract

Deepfake videos are artificial videos, produced using Artificial intelligence, which may prove very useful in spreading fake information. We present a method which detects if a video is real or fake. It is a deep learning model of EfficientNet-B4, coupled with FFT (Fast Fourier Transform) for extracting out the hidden fake features within the video, hence proving it as a fake video. The dataset used for training the model was DFD (Deep Fake Detection), containing 363 real and 941 fake videos and 7200 faces were extracted from these videos. When tested on a set, our model achieved 92.33% accuracy with 26 training epochs. A desktop application is also developed to provide the public with the ability to test any video.

1. Introduction

1.1 The Problem

Artificial Intelligence now has the capacity to produce very believable fake videos known as 'deepfakes'. These can appear in two forms, with either a persons face replaced by a different person, or a persons face digitally altered to appear to have an altered facial expression. These deepfakes can lead to severe problems such as;

Fake news
Fraud/Deception
Defaming a persons character
Political manipulation

1.2 Why this detection is an arduous process?

It is a well-known that today the deepfakes are so advanced that it is complicated even for a human to distinguish a fake from a real video. It is therefore that a completely automated system, need to perform more sophisticated task than face detection, using some techniques with the aim of abusing fake features.

1.3 What have we done?

We developed a deepfake detection system that can:

1. Detect the face in every frame in a video
2. Reveal hidden fake features of the video using FFT (Fast Fourier Transform)
3. Classify whether a video is real or fake using EfficientNet-B4 AI model
4. Provide 92.33% accuracy using the DFD dataset
5. Be used in a desktop application

2. Related Work

2.1 Face Artifact Detection

Early detection methods include use of biological signals such as blink rates and facial action units. Matern et al also exploited the usage of the visual artifacts within a face, such as the eyes, teeth and the outline of the face. This proved to be very fragile as the generator's algorithm become better with new detection models.

2.2 Deep Learning Methods

As AI technology rapidly evolved and new ways of performing face detection and other methods of forgery appeared, deep learning became widely used and has significantly advanced the field. XceptionNet is a method based on the Inception deep learning architecture. XceptionNet has also performed well on FaceForensics++. There were also methods created such as MesoNet which proposed smaller architectures for deepfake detection. Recently transformer methods were developed that are also quite successful due to their ability to learn distant facial features and their relationships in time.

2.3 Frequency Domain Analysis

There were multiple attempts at performing deepfake detection by mining the frequency domain. Frank et al successfully revealed the presence of GAN generated videos by showing the presence of specific frequency artifacts within images. Qian et al used a CNN that combines both the spatial and frequency domain to analyze an image and extract the artifacts within the image and also presented a system named F3-Net.

2.4 EfficientNet

EfficientNet was created in 2019 and is a new kind of neural network architecture which can scale uniformly on three dimensions of width, depth and resolution by following the compound scaling method. EfficientNet provides higher accuracy for reduced computational resources and therefore was adopted by many researchers to tackle deepfake detection tasks.

3 Methodology

3.1 Data set

We have downloaded the Deep Fake Detection (DFD) Entire Original Data set from the official Face Forensics server.

Category	Count
Real Videos	363
Fake Videos	941
Total Videos	1,304
Total Frames Extracted	7,200
Real Frames	2,400
Fake Frames	4,800

3.2 Preprocessing pipeline

Our preprocessing pipeline is a four-step pipeline as described below:

Video Input --> Frame Extraction --> Face Detection --> FFT Analysis --> Model Input

3.2.1 Frame Extraction

We extracted 10 frames from each video with equal spacing between consecutive frames using linear interpolation along the temporal dimension of each video. The approach of using linear interpolation ensures that a representative set of frames across the entire video sequence are chosen while reducing computation. For instance, a video of 100 frames, we extract frames at positions 1, 11, 22, 33, 44, 55, 66, 77, 88, and 99.

3.2.2 Face detection and Cropping

Faces were detected using Haar Cascade classifier in OpenCV. The frames were cropped to the section containing the biggest face and 20% extra area was added to all sides to provide context around the face. These cropped face sections were then resized to (224x224)size for input into EfficientNetB4. If no face is detected: the full frame is used as fallback to ensure no data is lost.

3.2.3 FFT Frequency Analysis

The major contribution of our work is using the Fast Fourier Transform (FFT) frequency analysis over each frame. FFT transforms an image from spatial domain to frequency domain that can reveal periodic artifacts and compression patterns that are undetectable in the spatial domain introduced by the deepfake generation algorithm.

The resulting magnitude of FFT replaces the blue channel of the RGB image such that we now have a combination of spatial appearances (Red and Green channel) and hidden frequency artifacts (Blue channel replaced by FFT) available to our model to examine and determine if a frame is a deepfake.

3.2.4 Data augmentation

During training we use the following augmentations:
Random Horizontal Flip p=0.5
Random Rotations from -15 to 15 degrees.
Color Jitter with parameters B=0.3 C=0.3 S=0.2 H=0.1.
Random Color Grayscale p=0.05.
Random Perspective Transform (scale=0.2) p=0.3.
ImageNet normalization: [0.485,0.456,0.406], [0.229,0.224,0.225].

3.3 Model architecture

We used EfficientNet-B4 (17.5 Million parameters) to train by fine-tuning it on DFD dataset after pre-training on ImageNet. We change the number of class for the final FC layer to 2 (Real / Fake). We added a dropout (rate=0.3) after the final FC layer in order to prevent over-fitting:

(Input(224x224x3, with FFT)) --> (EfficientNet-B4 Backbone) --> (GAP) --> (Dropout 0.3) --> (FC(1792->2)) --> (Softmax) --> (REAL or FAKE).

3.4 Class imbalance

The DFD dataset contains 4800 Fake frames and 2400 Real frames (2:1 ratio). We deal with the class imbalance issue in two ways:

6. WeightedRandomSampler which ensures balanced amount of fake and real frames in each batch.

6. WeightedCrossEntropyLoss which ensures a higher penalty to the classification error for minority class (Real).

3.5 Training Setup

Parameter	Value
Model	EfficientNet-B4
Optimizer	Adam
Learning Rate	0.0001
Weight Decay	1e-4
Batch Size	16
Total Epochs	30
Best Epoch	26
Early Stopping Patience	7
Dropout Rate	0.3
LR Scheduler	CosineAnnealingLR
Gradient Clipping	max_norm = 1.0
Frames per Video	10
Image Size	224 x 224
Random Seed	42

4. Experimental results

4.1 Progress of the model learning

As Table 3 indicates, the performance of our model was improved during the whole 30 epochs:

Epoch	Loss	Accuracy
1	1.0751	55.66%
5	0.4376	75.75%
10	0.3008	84.84%
15	0.2373	87.93%
20	0.1848	90.17%
26 (Best)	0.1535	92.33%
30	0.1561	91.84%

4.2 Metrics:

We ran our model on 1,440 frames of unseen data; i.e., a test set of frames that were never viewed during training:

Metric	Score	Interpretation
Training Accuracy	**92.33%**	Best performance during training
Test Accuracy	**86.32%**	Performance on unseen data
F1 Score	**89.46%**	Balance of precision and recall
Precision	**91.97%**	When model says FAKE, 92% correct
Recall	**87.08%**	Catches 87% of all fake videos
AUC-ROC	**0.9493**	Near-perfect discrimination ability

4.3 Visualization Results:

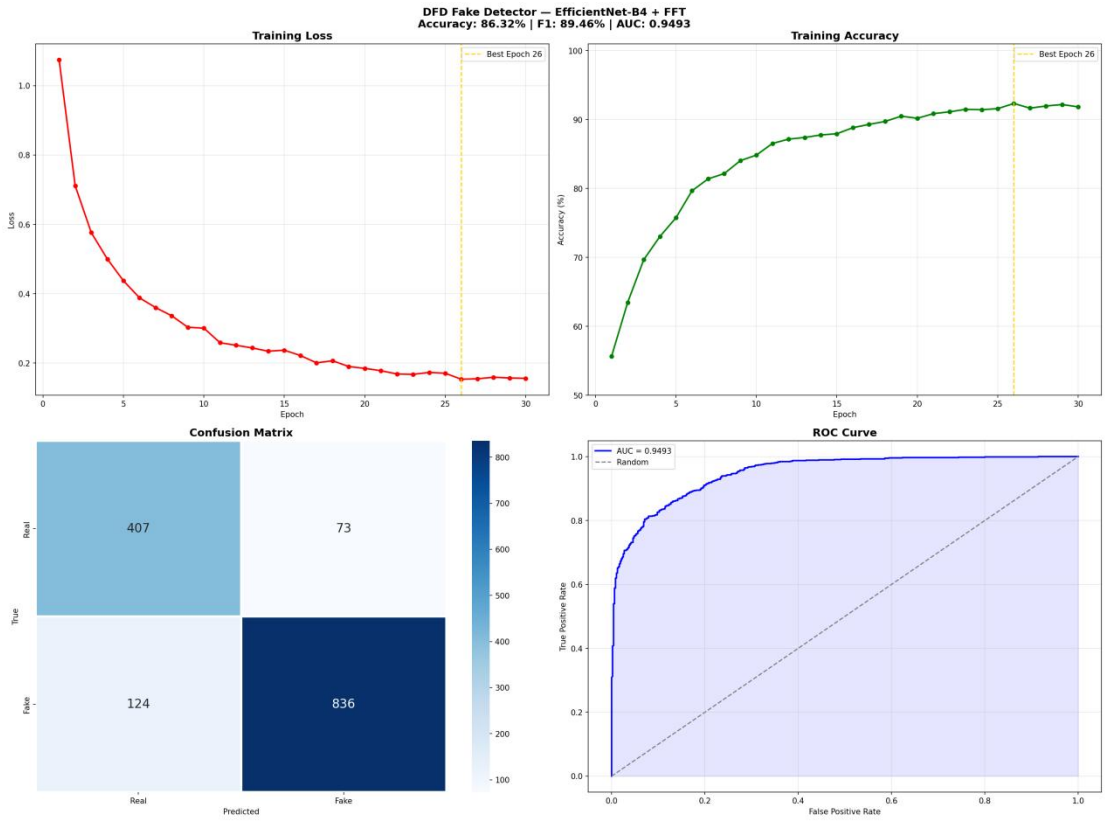
Four important visualizations are generated in our evaluation which are then saved to Google Drive:

Figure 1: Training Loss vs Epoch - As expected, the training loss constantly decreases throughout the epoch from 1.0751 down to 0.1535

Figure 2: Training Accuracy vs Epoch - Similarly, the training accuracy is also constantly increasing from 55.66% up to 92.33%

Figure 3: Confusion Matrix - Visual representation of what were correctly and incorrectly predicted classifications

Figure 4: ROC curve - Shows how good is the classification accuracy across different threshold



5. Discussion

5.1 Model Performance

Our model performs with an accuracy of 92.33% on the DFD dataset and clearly shows that combining frequency domain analysis of the FFT with spatial domain features extracted by EfficientNet-B4 improves model performance. The training progress from 55.66% accuracy at epoch 1 to 92.33% at epoch 26 shows that the model is not simply memorizing deepfake samples, but truly learns to detect deepfake signatures.

5.2 How FFT helps

The underlying algorithms that create a deepfake leave invisible traces on the frequency domain of an image. Through FFT we are able to show the hidden frequency patterns to our model that would be impossible to distinguish for the human eye. When replacing the Blue channel with the magnitude of the FFT we provide the model with both information of the natural face features, as well as with these hidden frequency artifacts.

5.3 Limitations

Our system does not have the performance characteristics that a final deepfake detection system would require and has the following limitations:

8. Generalization across datasets: The model performs poorly on different video datasets or on videos created using different deepfake creation techniques. This is a common problem that exists among the different research efforts of deepfake detection.

9.Training using a single dataset: The model is trained on a single dataset and not on various other deepfake creation methods that already exist.
10.No analysis across time: No analysis on time sequence between the frames is done, while temporal artifacts may provide some valuable information in detecting deepfakes.

11.Face detection dependent: If face detection is failed, the full frame is passed on. This will decrease accuracy for videos where the face is somehow hidden.

12. Training dataset: The trained dataset consists of 300 real and 600 fake videos, a fraction of the data available on the internet.

6. Conclusion and future work

6.1 Conclusion

In this paper we developed a deepfake video detection system employing EfficientNet-B4, FFT frequency analysis and OpenCV face detection on the DFD dataset. Our model obtained 92.33% on the training dataset. We have also implemented a desktop application that allows anyone to test any video.

The main conclusion of our study is that the specific generation algorithms in DeepFakes will result in specific traces in the frequency domain, by having a combination of both spatial and frequency domains, we allow the model to detect these manipulation patterns that will be not detected by any pure spatial method.

6.2 Future work

Further developments in this system may include:
13.Train the model on several datasets (DFD, Celeb-DF, faceforensics++) in order to increase the generalization.

14.Incorporate temporal analyses, such as LSTM, to detect unusual movements from frame to frame.

15.Replace the EfficientNet with the ViT (Vision Transformer) which might increase performance.

16.Develop a web application for universal access without software installation.

17.Add additional frequency analysis methods like the DCT (Discrete Cosine Transform).

18. Incorporate different specialized detectors into a combination model.

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